



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE AND INFORMATION TECHNOLOGY DEPARTMENT
BS (COMPUTER SCIENCE)

COURSE INFORMATION SHEET

Session:	Spring-2024
Course Title:	Database Systems
Course Code:	CS-229
Credit Hours:	3
Pre-Requisites:	None
Instructor Name:	Dr. Anam Akbar Ms. Anam Siddiqui
Email and Contact Information:	ansiddiqui@ssuet.edu.pk , HS-16, 34973414 Ext: 315
WhatsApp Group	Database-24
Office Hours:	Mon: 9:00 am – 1:00 pm
Mode of Teaching:	✓ Synchronous/Asynchronous/Hybrid/Blended

COURSE OBJECTIVE:

The objectives of this Course are

- List and explain the fundamental concepts of a relational database system..
- Analyze database requirements and determine the entities involved in the system and their relationship to one another.
- Develop the logical design of the database using data modeling concepts such as entity-relationship diagrams.
- Create a relational database using a relational database package.
- Manipulate a database using SQL.
- Assess the quality and ease of use of data modeling and diagramming tools

COURSE OUTLINE:

Basic database concepts; Entity Relationship modeling, Relational data model and Relational algebra, Structured Query Language; RDBMS; Database design, functional dependencies and normal forms; Transaction processing and optimization concepts; concurrency control and recovery techniques; Database security and authorization. Database efficiency and tuning; Data warehousing and Data Mining, Emerging Database Technologies and Applications.



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COURSE LEARNING OUTCOMES:

CLO No.	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom's Taxonomy
CLO 1	Describe concepts, theories, and processes in database systems.	PLO_1 (Academic Education)	C1 (Knowledge)
CLO 2	Identify and Illustrate the factors that lead to the development and implementation of Database systems.	PLO_2 (Knowledge for Computing Problem)	C2 (Understanding)
CLO 3	Apply practices of designing in database systems.	PLO_3 (Problem Analysis)	C3 (Applying)

RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO1	CLO2	CLO3
Quizzes	3 (9.6%)	3 (8.3%)	4 (12%)
Assignments	3 (9.6%)	3 (8.3%)	4 (12%)
Mid	15(48.3%)	15(41.6)	-----
Final	10(32.2%)	15(41.6%)	25(76%)
Total	31 (31%)	36 (26%)	33 (33%)

GRADING POLICY:

Assessment Tools	Percentage
Quizzes	10%
Assignments	10%
Midterm Exam	30%
Final Exam	50%
TOTAL	100%

Recommended Book:

- Elmasri, Ramez, *Fundamentals of Database Systems*, 7th Edition, Pearson India, 2017

Reference Books:

- C.J.Date: *An Introduction to Database Systems*, 8th Edition
- Coronel, Carlos, *Database Systems: Design, Implementation, & Management*, 13th Edition Cengage Learning, 2018



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- Building the Data Warehouse W.H. Inmon, Wiley Publishing Inc., Latest Edition.
Inmon, Building the Data Warehouse, 4th edition, Indiana, Wiley Publishing Inc., 2005
- Data Mining: Concepts and Techniques Book by Jiawei Han Jiawei, Data Mining: Concepts and Techniques, 3rd Edition, San Francisco, Elsevier Inc., 2012

Relevant Job:

- Oracle Developer
- Database Business Administrator

Benchmarking:

- University Of Warwick - UK

<https://warwick.ac.uk/fac/sci/dcs/teaching/modules/cs258/>

- Bahria University -PAK

<https://www.bahria.edu.pk/wp-content/uploads/2017/06/Database-Management-Systems.pdf>



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LECTURE PLAN

Course Title: Database System

Course Code: CS -229

Week No.	Week Dates	Topics	Required Reading	Key Date
1	12-03-2024 to 15-03-2024	Introduction to the Database, what is Data, Information, and Database Information Systems, Actors on the Scene, Workers behind the scene, Advantages of using Database approach, Data Models, Schemas, and Instances, Three-Schema Architecture	RE-Chap 1: Pg. 4 RE-Chap 1: Pg. 17 RE-Chap 2: Pg. 32	
2	19-03-2024 to 22-03-2024	- Data Independence - Centralized and Client/Server Architectures for DBMS - Database security - Database Authentication - Database auditing	RE-Chap 2: Pg. 36,46 CJ-Chap 2 Pg. 49 RE-Chap 30: Pg.1121 RE-Chap 30: Pg.1126	
3	26-03-2024 to 29-03-2024	- Using High-Level Conceptual Data Models for Database Design - Data Independence - Relationship Types, Relationship Sets, Roles, and Structural Constraints	RE-Chap 2: Pg. 31 RE-Chap 2: Pg. 36 RE-Chap 3: Pg. 63,72	Quiz # 1
4	01-04-2024 to 05-04-2024	- Weak Entity Types, Total and Partial Participation - Existence Dependencies - Associative Entity sets, Unary, Binary, Ternary relationship sets, Recursive relationship	RE-Chap 3: Pg. 72-79	Assignment #1
5	08-04-2024 to 12-04-2024	- Subclasses, Superclasses, and Inheritance - Specialization and Generalization - Constraints and Characteristics of Specialization and Generalization Hierarchies	RE-Chap 4: Pg. 107-113	



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6	15-04-2024 to 19-04-2024	- Relational Model Concepts - Constraints and Relational Database Schemas - Update Operations, Transactions, and Dealing with Constraint violation	RE-Chap 9: Pg. 290,298 RE-Chap 5: Pg. 165	
7	22-04-2024 to 26-04-2024	- Relational Database Design Using ER-to-Relational Mapping	RE-Chap 9: Pg. 290,298	Quiz # 2
8	29-04-2024 to 03-05-2024	- Mapping EER Model Constructs to Relations - Mapping Super/Sub sets relationships, and Shared Subclasses	RE-Chap 9: Pg. 301	Assignment #2
9	Midterm Examination (06-05-2024 to 11-05-2024)			
10	13-05-2024 to 17-05-2024	- SQL Data Definition and Data Types - INSERT, DELETE, and UPDATE Statements in SQL - Basic Retrieval Queries and Constraints in SQL, Exercises for SQL - Basic tuning concepts	RE-Chap 6: Pg. 179 CJ-Chap 4 Pg. 85 RE-Chap 6: Pg. 198,201 RE-Chap 6: Pg. 187,205	
11	20-05-2024 to 24-05-2024	- Basic SQL Relational Algebra Operations - Unary Relational Operations - Relational Algebra Operations From Set Theory - Binary Relational Operations	RE-Chap 8: Pg. 239-251	
12	27-05-2024 to 31-05-2024	- Functional Dependencies and its types - Normal Forms Based on Primary Keys - General Definitions of first ,Second and Third Normal Forms, Exercises For Normalization,	RE-Chap 15: Pg. 503 RE-Chap 14: Pg.471,474 CJ-Chap 11 Pg. 33,349 RE-Chap 14: Pg. 483,497 CJ-Chap 12 Pg. 357	



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13	03-06-2024 to 07-06-2024	- Introduction to Transaction Processing Concepts and Theory - Desirable Properties of Transactions - Two-Phase Locking Techniques for Concurrency Control	RE-Chap 20: Pg.745 RE-Chap 20: Pg.757 RE-Chap 21: Pg. 782	Quiz # 3
14	10-06-2024 to 14-06-2024	- Multiversion Concurrency Control Techniques - Recovery Concepts,	RE-Chap 21: Pg. 795 CJ-Chap 16 Pg. 465 RE-Chap 21: Pg. 798,800 RE-Chap 22: Pg. 814	Assignment #3
15	17-06-2024 to 21-06-2024	- Granularity of Data Items - Multiple Granularity Locking - Concurrency Control Issue	RE-Chap 21: Pg. 798,800 CJ-Chap 16 Pg. 466 RE-Chap 21: Pg. 798,800 CJ-Chap 16 Pg. 466	
16	24-06-2024 to 28-06-2024	- Introduction Data Warehousing and Data Mining	WH - Chap:1 Page. N0.1-25 WH - Chap:1 Page. N0.357	
17	01-07-2024 to 05-07-2024	- Emerging Database Technologies and Applications	Research Papers and Latest articles RE-Chap 25: Pg. 911	
18	08-07-2024 to 12-07-2024	- Big Data	Research Papers and Latest articles RE-Chap 25: Pg. 911	
Final Examination (15-07-2024 to 26-07-2024)				

Name and Signature (With Date): Anam Siddiqui 21-03-2024

Chairperson's Signature of DoC&IT: _____